Graph-based machine learning

or Networks

Tabular data

- A sample is defined based on common features
- Independent of others
- Shared features

	Feature 1	Feature 2
Sample 1		
Sample 2		

Limits of tabular data

- What defines an influencer?
- What defines a route?
- What defines a concept?

Network science

- Samples defined by its relations with other samples
- Samples & ~~features~~ samples
  - Collaborative definition of knowledge (dictionary)
  - NxN
- Nodes/**V**ertices
- Links/**E**dges

	Sample 1	Sample 2
Sample 1		
Sample 2		

Graph Theory

- $G=(V,E)$, $E \subseteq V \times V$
  - $e \in E$, $(v,w) \in V^2$, v and w are adjacent
- Complete graph / Clique
- Subgraph H of graph G
  - $V(H) \subseteq V(G)$, $E(H) \subseteq E(G)$

Vertex properties

- Vertex **degree**
- Neighbourhood / **Ego network** / N-step neighbourhood
- A **walk** $w = (v_1, v_2, \ldots, v_n) \in V$ where consecutive vertices are adjacent
- A **path** is a walk where no vertex is visited twice
- **distance(v,w)** = length of shortest path

Vertex connectivity

- (v_1,v_2) are **connected** if there is a path between them
- Connected vertices define graph **connected components**
  - **Connected graph** = Single connected component
  - Disconnected components beak everything
- **Bi-partite graph**: Two sets of vertices with no internal connection

Relation types

- **Existence** (adjacency matrix/list)
  - Sparse matrix representations
  - e.g., dictionary of keys: (1,1), (2,1), (3,2)

	Sample 1	Sample 2	Sample 3
Sample 1	X		
Sample 2	X		
Sample 3		x	

- **Undirected/Directed**
  - Triangular matrix
  - Vertex Indegree/Outdegree

Relation types

- **Weighted**
  - Numbered edges
- **Labeled**
  - Named edges
- **Cyclic/Acyclic**
  - Loops (diagonal)
  - Trees are connected, acyclic graphs

	Sample 1	Sample 2	Sample 3
Sample 1	2		
Sample 2	1		
Sample 3		0.6	

	Sample 1	Sample 2	Sample 3
Sample 1	self		
Sample 2	subtype		
Sample 3		part of	

Traversing graphs

- Data access is not linear
  - Data-driven
  - Foundation of most algorithms
- ~~Locality~~
- ~~Predictability~~
- Memory issues

Graphs scope

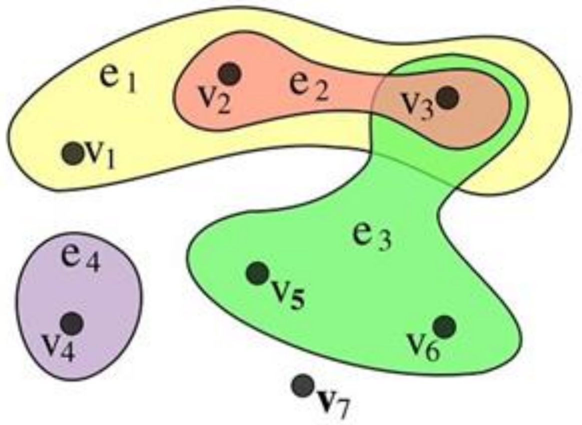
- Local
  - Ego net, Degree
- Quasi-local
  - Walks
- Global
  - Adjacencies, centrality, etc.

Graphs 2.0

Extending networks

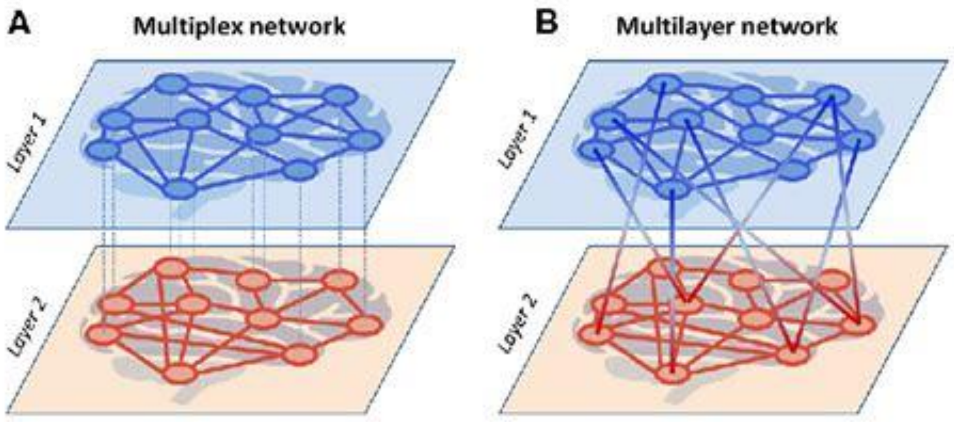
Fancy graphs

- Hypergraphs
  - Hyperedges connect one or more vertices
  - Multi-entity interactions
  - Connecting with connections



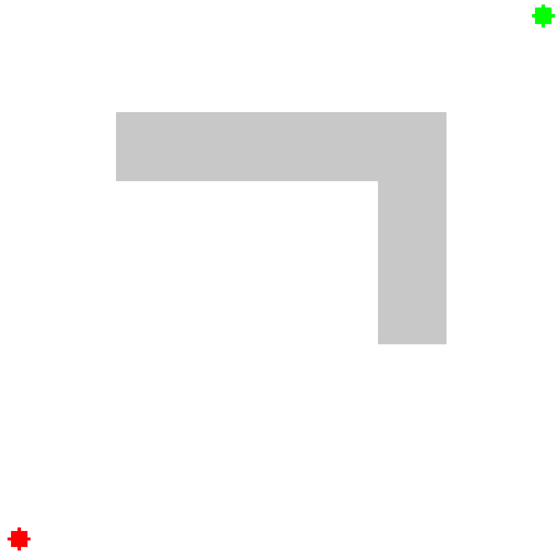
Fancy graphs

- Multilayer graphs
  - Edges across layers
- Multiplex
  - Edges across layers only with self



Path finding

- Dijkstra algorithm
  - Shortest path between two vertices
  - Choose next one based on total cost
  - All paths found in cost order



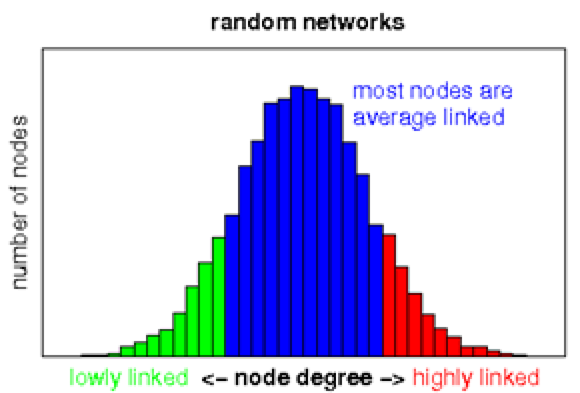
- ABIA search algorithms
  - BFS, DFS, A* …

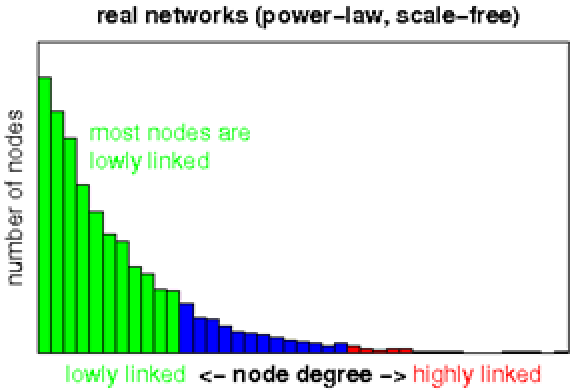
Graph properties

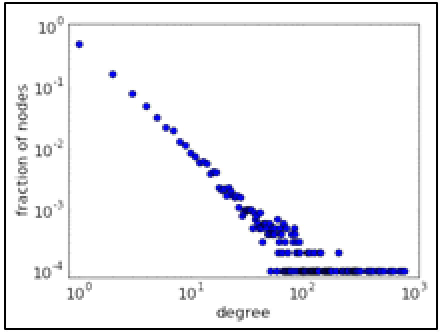
- Local
  - Node degree
    - Hubs (out-degree)
    - Authorities (in-degree)

Graph properties

- Global
  - Degree distribution
  - Density = |E|/((|V|*|V|-1)/2)
  - Diameter: longest shortest path







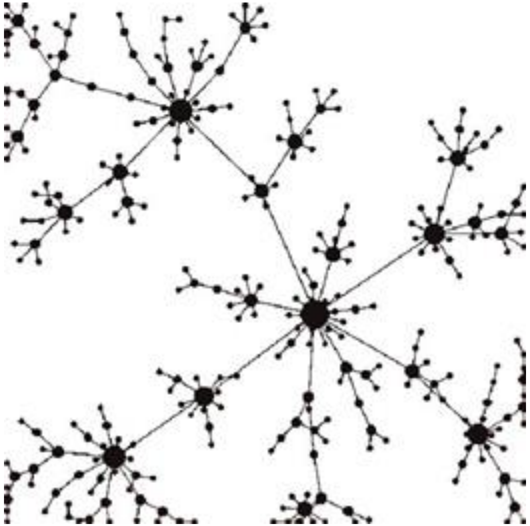
A node scope

- Local
  - Ego net, Degree
- Quasi-local
  - Walks
- Global
  - Adjacencies, centrality, etc.

Purpose & size!

Preferential Attachment

- *“Rich get richer”*
  - Probability of new edge depends on degree
  - Power-law distributions
  - Scale-free graphs
  - Everywhere



Graph paradox

- Graphs are often sparse
  - Possible E quadratic
  - Actual E linear
- And yet, diameters are typically small
  - Six Degrees of ~~separation~~ Kevin Bacon
  - Hubs & Authorities

Graph stuff

What can we do

Graphs can be rich

- Topology alone is limited
- Expanding Nodes/Edges
  - Labels, weights or attributes
- More specific, more performant, less general

Possibilities

- Rank nodes
- Compare nodes
- Find missing edges
- Cluster nodes
- Find frequent structures

Centrality measures

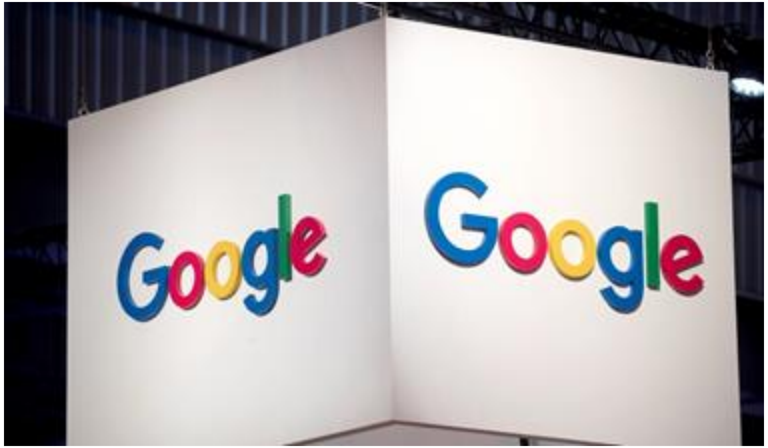
And link-based object ranking

Node centrality

- Degree: Num. edges
  - Audience
- Closeness: Avg len. of shortest path with all other nodes
  - Accessibility
- Betweenness: Num. of shortest paths which pass through it
  - Criticality
- …

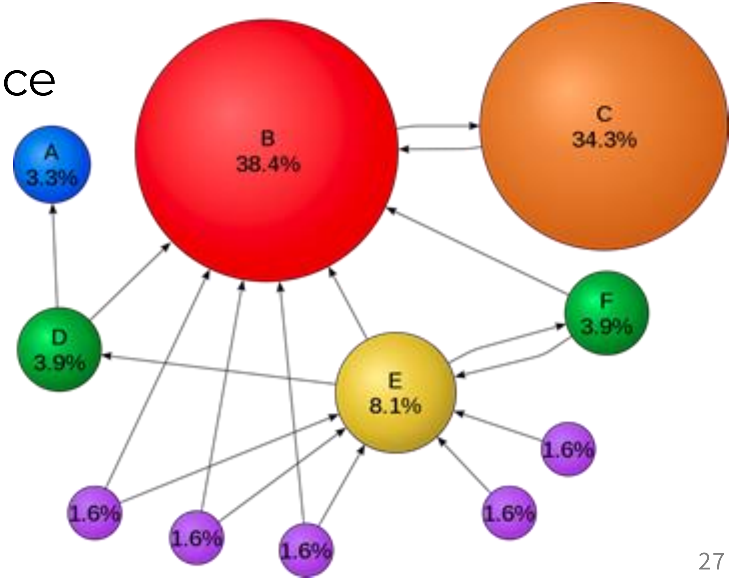
Page Rank: Topological relevance

- All vertices/edges are not equally important
- Importance based on connectivity
- Relevance flows through directionality
- Perfect for ranking webpages
  - Hubs (directories)
  - Authorities (wikipedia)



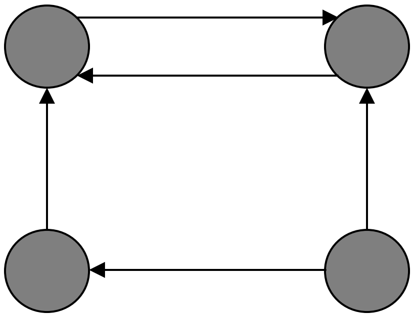
Page Rank: Relevance is contagious

- Measure probability of random walk end
  - Assign same relevance to all
  - Each vertex equally distributes relevance among (out) neighbours
  - Iterate until convergence

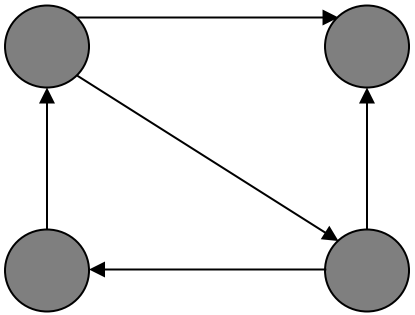


Page Rank: Relevance is contagious

- Spider traps
  - No links outside group



- Sink nodes
  - Node no out

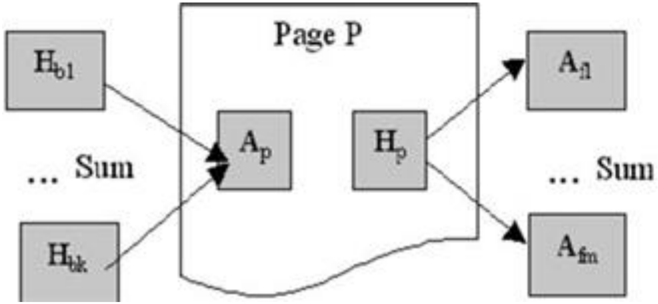


Teleport

- Avoid spider traps and sink nodes
    - Random jump with low prob.
    - Complete graph
    - “damping factor”

HITS

- Two scores per node
  - In-degree: Authority(v) = ∑(Hubs(u) ∀u where (u,v)∈E
  - Out-degree: Hub(v) = ∑(Authority(u) ∀u where (v,u)∈E
  - Init all to constant
  - Compute new values
  - Normalize
  - Iterate until convergence



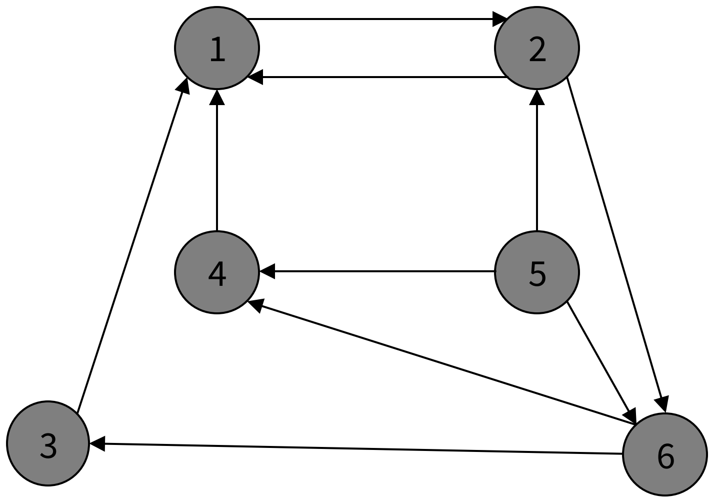
Graphs by hand

- HITS
  - A(v) & H(v) = 1
  - Compute new values

Authority(v) = ∑(Hubs(u) ∀u where (u,v)∈E

Hub(v) = ∑(Authority(u) ∀u where (v,u)∈E

  - Iterate until convergence



Node similarity

A distance measure gets you far

Accumulative Measures (sample)

- Common Neighbours

$$ScoreCN(u,v) = |\Gamma(u) \cap \Gamma(v)|$$

  - Absolute neighbour overlap

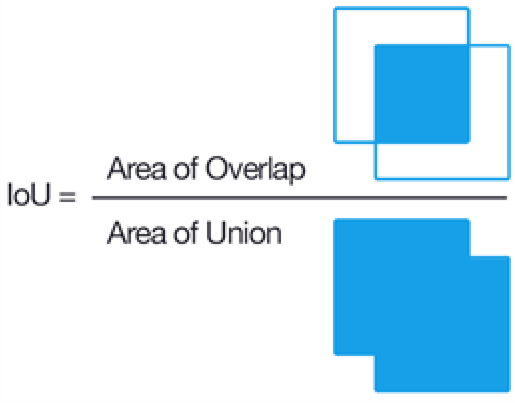
- Adamic Adar, Resource Allocation
  - Absolute neighbour overlap, weighted by their own degree

$$ScoreAA(u,v) = \sum_{z \epsilon \Gamma(u) \cap \Gamma(v)} \frac{1}{\log |\Gamma(z)|}$$

Proportional Measures (sample)

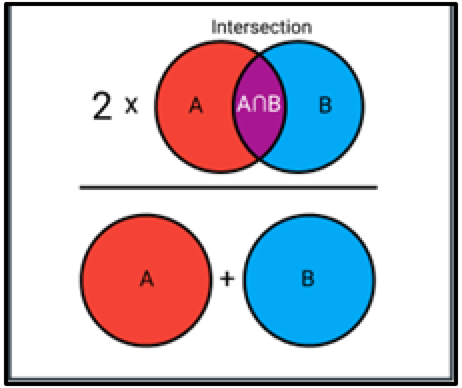
- Jaccard index aka IoU

$$ScoreJC(u,v) = \frac{|\Gamma(u) \cap \Gamma(v)|}{|\Gamma(u) \cup \Gamma(v)|}$$



- Sørensen–Dice coefficient

$$ScoreSO(u,v) = \frac{2 * |\Gamma(u) \cap \Gamma(v)|}{\Gamma(u) + \Gamma(v)}$$



Local vs Quasi-local vs Global

- Local
  - e.g., CN
- Quasi-local
  - e.g., random walks of len k)
- Global
  - e.g., Katz (num. shared paths)

Link Prediction

Finding your soulmate

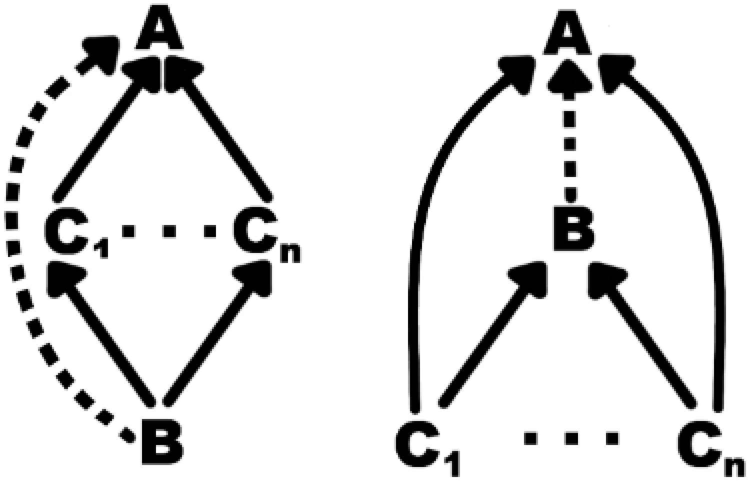
As a binary classification problem

- Each edge holds little value on it’s own
  - Predictive power vs noise

- Edges that exist vs Edges that don’t
  - Imbalanced, thx to PA and size
  - Up/Downsampling not a good idea

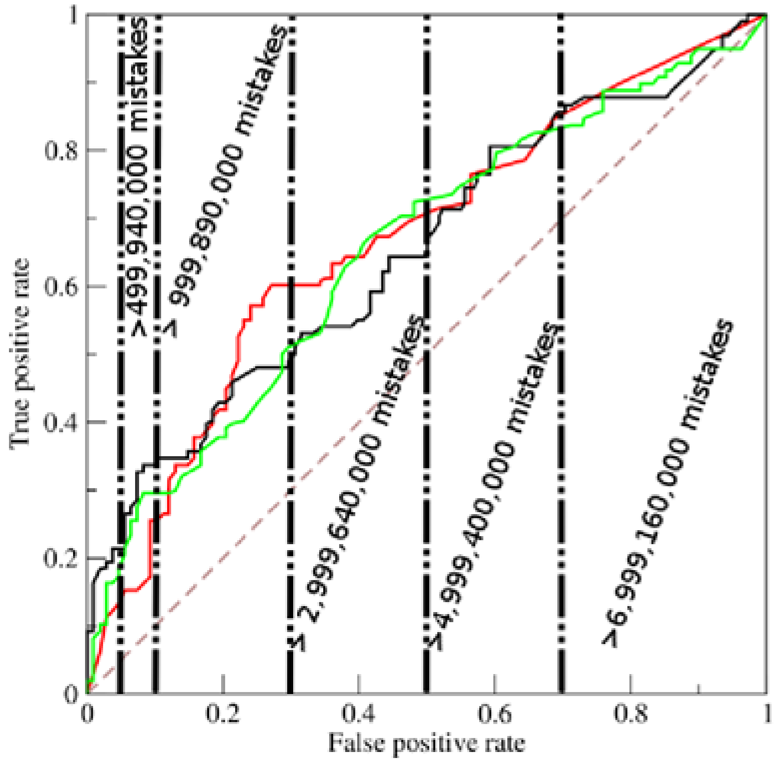
Candidates for Link Prediction

- Order and join by node similarity
- Exploit properties
  - e.g., hierarchy & inheritance



LP breaks evaluation metrics

- N is too big (wrt P)
  - Distorted ROC curve
  - TN need exponential scale
- ~~ROC~~ Precision-Recall curve
  - High recall impossible



|V|=100K |E|=1M 90/10 split

|P|=100K |N|= 9998.8M

Community Detection

Clustering graph-style

Unsupervised

- No ground truth
- No universal metrics
- Synthetic graphs are synthetic
  - Different graph models
  - No free lunch theorem
- Diversity as a value

And yet… we evaluate

- Normalized mutual information (X,Y)
  - Expected and actual communities
  - Measure of info about X given by Y

- Modularity (X)
  - Fraction of edges within group - expected fraction at random
  - Higher, more distinct cluster. Lower, more diffuse cluster

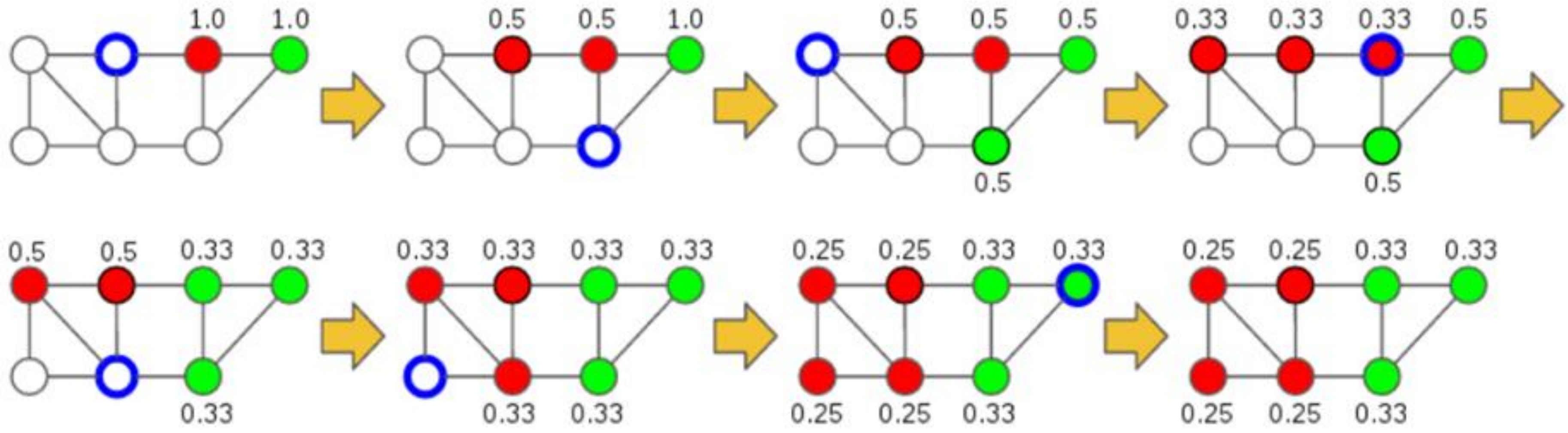
Basic approaches

- Similarity-based + Hierarchical
  - Heuristic or topological
- Minimum cut
  - Requires k. Expensive
- Betweenness centrality (e.g., Girvan–Newman)
  - Remove edges with high BC (bottlenecks). Expensive

Basic approaches

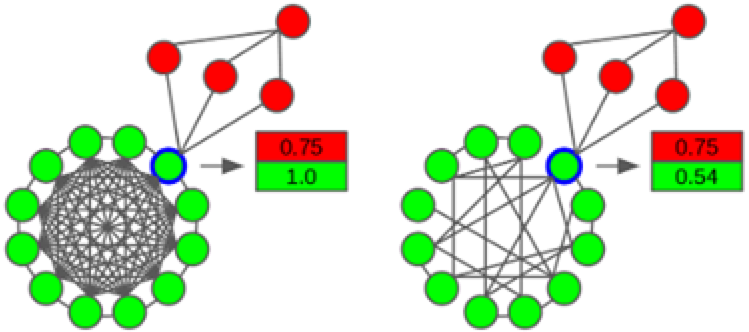
- Modularity maximization (e.g., Louvain) *Expensive*
  - Biased towards large communities.
- Label propagation (e.g., LPA) *Very cheap!*
  - Assume most frequent neighbour label
  - Run until convergence (asynchronous! non-deterministic!)
  - Initialize distinctly, propagate in random order. Equal prob. on ties
  - Boosts with semi-supervision and similarity heuristics

Fluid Communities (Label propagation + Density)



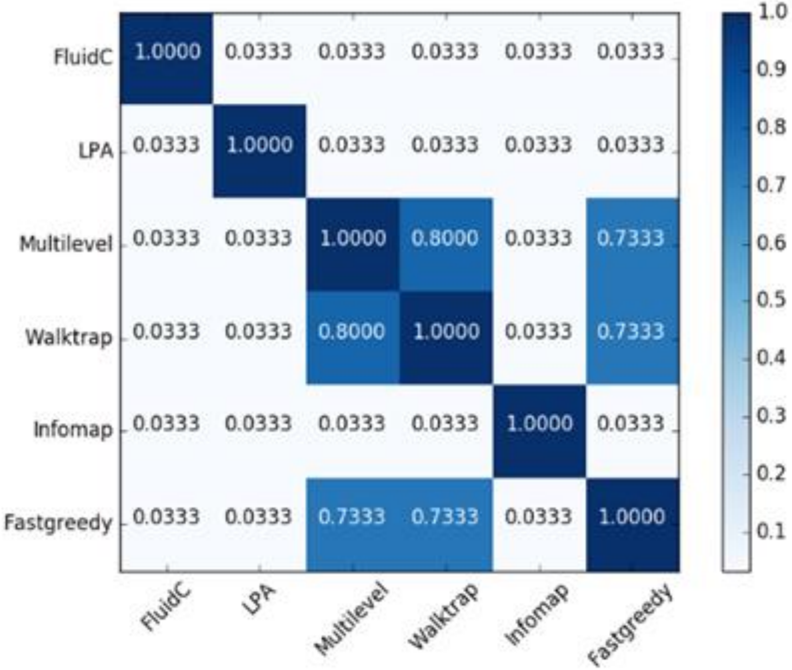
Fluid Communities

- Holding bottlenecks
    - Density & Uniformity
- Impossible to kill a community
- Balances communities sizes



How to choose

- This is clustering
  - Domain knowledge
  - Algorithm independent information
    - Variance and exploration



Frequent Subgraph mining

Subgraph discovery

Domain specific

- Some general approaches
  - Apriori on adj. matrix
    - Vertices are both, transactions and itemsets
  - Characterize subgraphs based on frequent sub-subgraphs